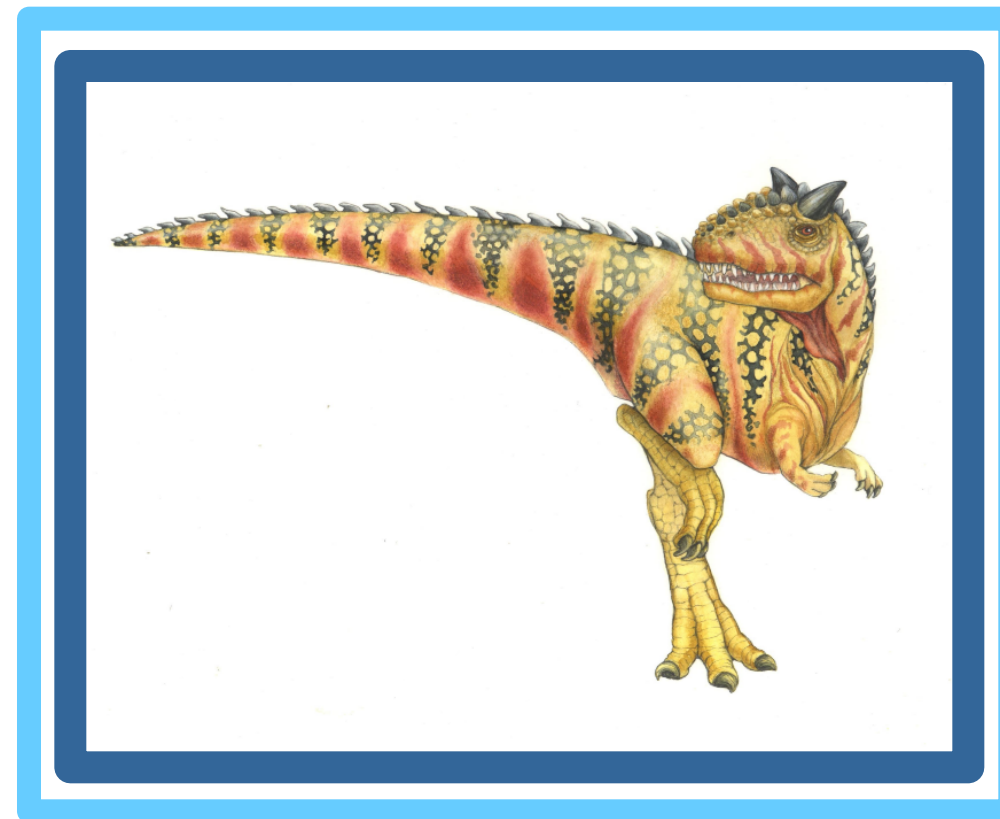
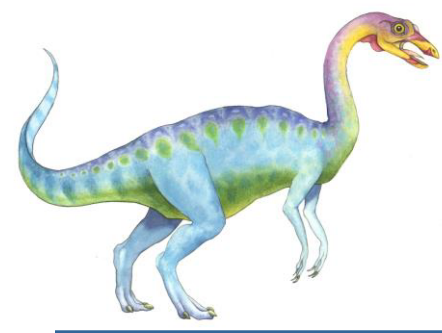


Chapter 8: Deadlocks





Chapter 8: Deadlocks

- ☐ System Model
- ☐ Deadlock Characterization
- ☐ Methods for Handling Deadlocks
- ☐ Deadlock Prevention
- ☐ Deadlock Avoidance
- ☐ Deadlock Detection
- ☐ Recovery from Deadlock





Deadlock Detection

Allow system to enter deadlock state

Detection algorithm

Recovery scheme





Single Instance of Each Resource Type

Maintain **wait-for** graph

Nodes are processes

$P_i \rightarrow P_j$ if P_i is waiting for P_j

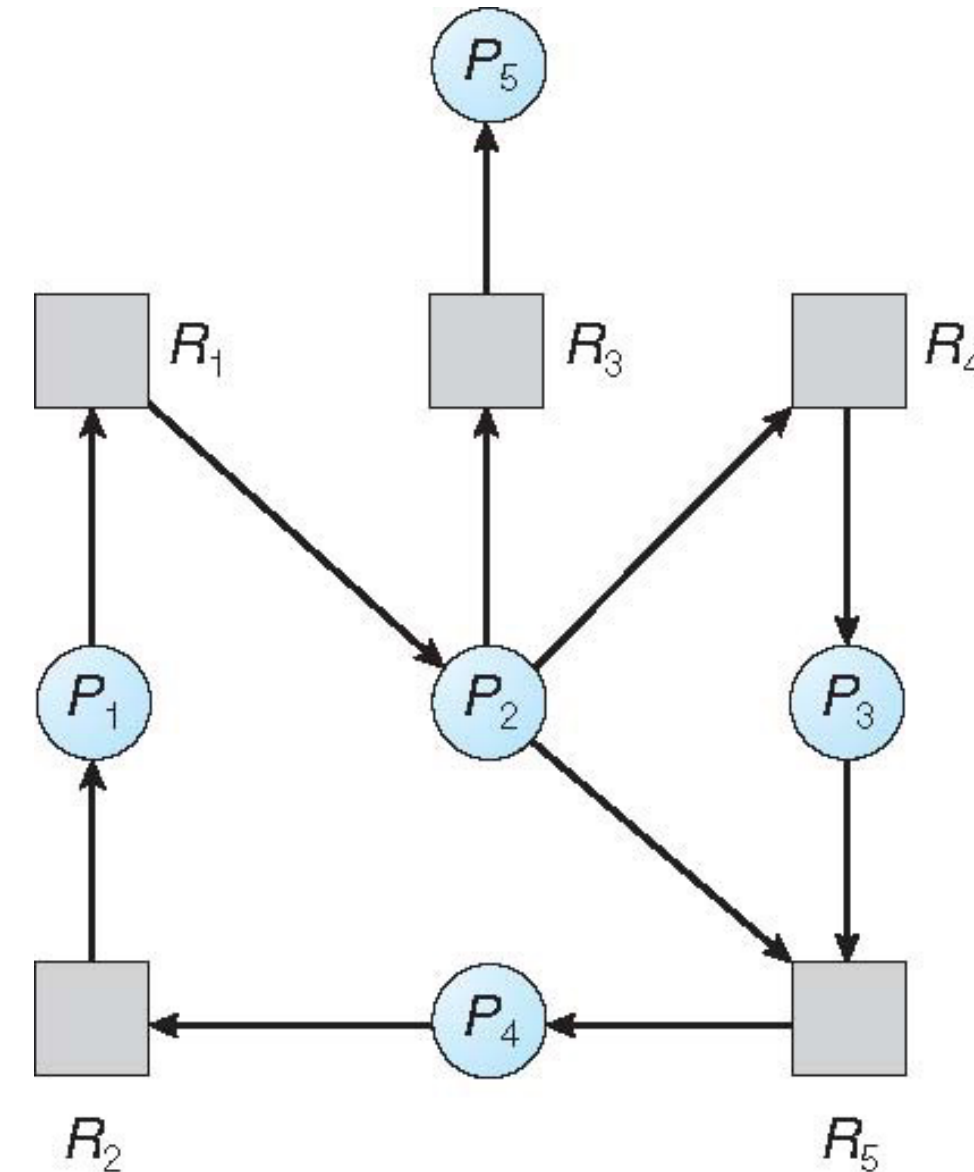
Periodically invoke an algorithm that searches for a cycle in the graph.
If there is a cycle, there exists a deadlock

An algorithm to detect a cycle in a graph requires an order of n^2 operations, where n is the number of vertices in the graph



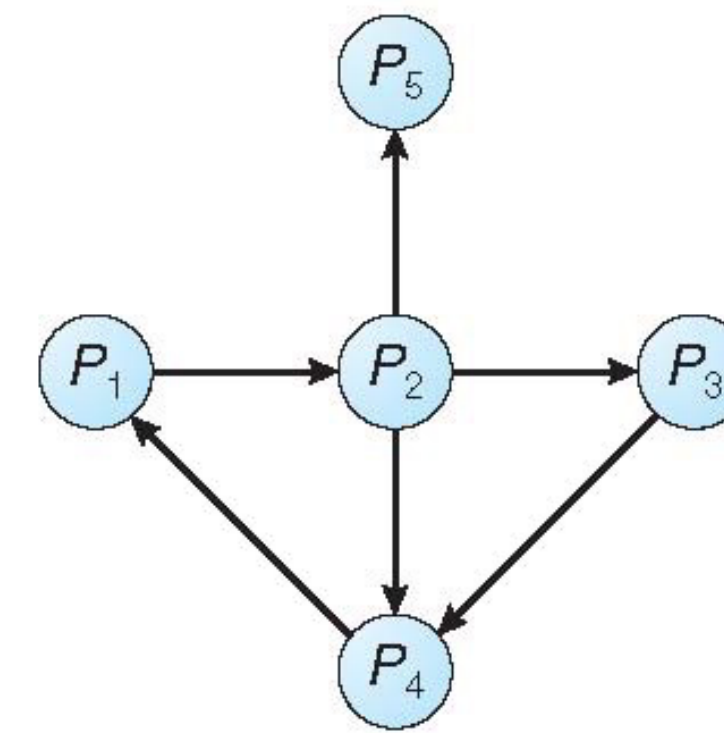


Resource-Allocation Graph and Wait-for Graph



(a)

Resource-Allocation Graph



(b)

Corresponding wait-for graph





Several Instances of a Resource Type

Available: A vector of length m indicates the number of available resources of each type

Allocation: An $n \times m$ matrix defines the number of resources of each type currently allocated to each process

Request: An $n \times m$ matrix indicates the current request of each process. If $\text{Request}[i][j] = k$, then process P_i is requesting k more instances of resource type R_j .

Banker's Algorithm





Recovery from Deadlock: Process Termination

Abort all deadlocked processes

Abort one process at a time until the deadlock cycle is eliminated

In which order should we choose to abort?

1. Priority of the process
2. How long process has computed, and how much longer to completion
3. Resources the process has used
4. Resources process needs to complete
5. How many processes will need to be terminated
6. Is process interactive or not?





Recovery from Deadlock: Resource Preemption

Selecting a victim – minimize cost

Rollback – return to some safe state, restart process for that state

Starvation – same process may always be picked as victim, include number of rollback in cost factor



Quiz

1. What are the 4 approaches to handle deadlocks?
2. What are the 4 approaches for Deadlock prevention?
3. What are the 2 approaches for Deadlock Avoidance?
4. What are the 2 approaches for Deadlock Detection?
5. What are some of the criteria to select a victim process?
6. What is the main difference between resource allocation graph and wait for graph?

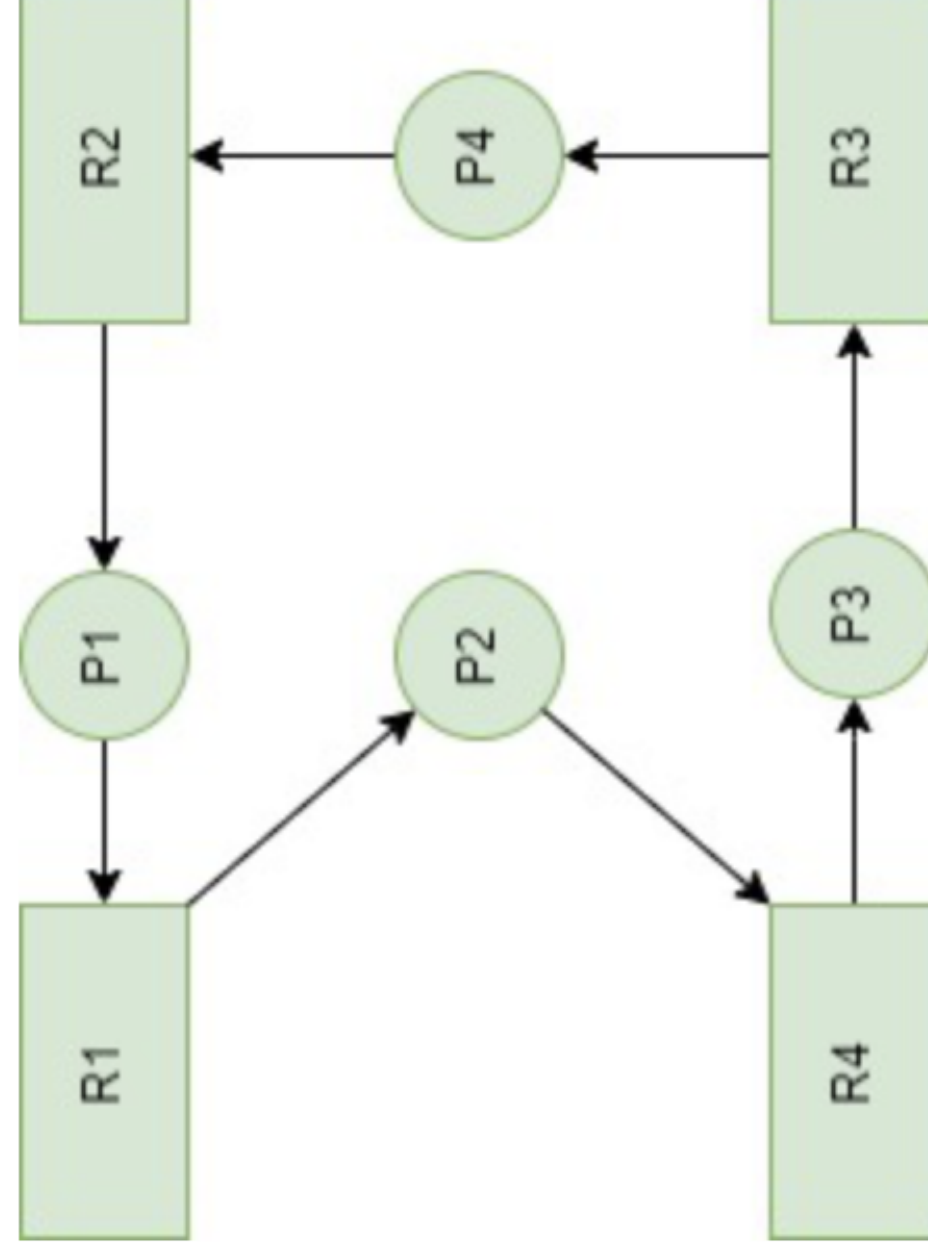


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System Model
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RAG to Wait for Graph



Wait for Graph

